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AI-Driven Academic Advising in Higher Education: Leveraging Intelligent Systems to Personalize Student Support, Improve Retention, and Optimize Career Pathways

Aftab Ahmed Soomro[1], Muhammad Hayat Khan[2], Muhammad Umar[3], Sajid Khan[4], Dr. Osama Ali[5]

[1]*Department of Mechanical Engineering Technology, Benazir Bhutto Shaheed University of Technology & Skill Development, Khairpur Mirs, Pakistan, Email: aftabsoomro@bbsutsd.edu.pk*
[2]*PhD Education, GIFT University Gujranwala, Email: m.hayat@gift.edu.pk*
[3]*Department of Management & Social Sciences, Abasyn University, Islamabad Campus, Pakistan, Email: mumar9400@gmail.com*
[4]*Department of Mechanical Engineering Technology, Benazir Bhutto Shaheed University of Technology & Skill Development, Khairpur Mirs, Pakistan, Email: sajidkhan@bbsutsd.edu.pk*
[5]*Islamic University of Madina, Email: oa02226@gmail.com*

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Corresponding Author:
Muhammad Hayat Khan
Email:
m.hayat@gift.edu.pk

ABSTRACT

The integration of Artificial Intelligence (AI) into academic advising is rapidly transforming how higher education institutions support student success. This study explores the effectiveness of AI-driven academic advising systems in enhancing student support, improving retention rates, and optimizing career pathways. Using a survey-based methodology, data was collected from 1,200 university students across three institutions that have implemented AI-powered advising platforms. The survey assessed students' experiences with the AI system, focusing on usability, personalization, academic confidence, retention likelihood, and alignment of academic choices with career goals. Results indicate a significant positive impact of AI on students' perception of guidance quality, timely decision-making, and clarity in career planning. Furthermore, students reported increased satisfaction and a stronger sense of being supported academically, with many noting that AI-generated insights helped them make informed choices about courses and careers. These findings suggest that when implemented ethically and with proper institutional support, AI-driven academic advising can serve as a powerful tool for personalized education and long-term academic success.

1. Introduction

1.1 Background

Historically across different levels of learning, academic advising has been an important student support system that assists learners in directing and providing them with the necessary information in course choice, career directions, and self-actualization. Academic advising practices have involved physical meetings between the student and the advisor, where the advisor has to use their experience, as well as the student's records to guide the student. However, over the years as the educational systems continue to experience changes and integration of technology that challenges the manner in which academic institutions provide services, the model of academic advising has been stretched. The large student enrollment, scarcity of advisors, and variability in students' needs are some of the reasons why it may be challenging and time-consuming for a human advisor to attend every student effectively [1]. Therefore, more centers are seeking technological solutions where AI is seen as an innovative technique that can complement or strengthen the academic advising process [2].

Machine learning, NLP and data analysis are some of the subfields of AI, and they have impacted many industries through automation of tasks, quick decision making and offering customized suggestions. Regarding HE, AI-based academic advising solutions are being designed to optimize advising, make it more personalized and scalable [3]. They consume and assess large amounts of data in the students such as performance, classes taken, clubs and organizations joined and even career preferences in order to make recommendations and provide timely information to those students. Initial work and adoption demonstrate that AI-based advising systems enhance overall student retention, attendance rate, and career match, which are all essential components of student achievement [4].

Thus, the use of AI in college advising also has its benefits and drawbacks. On the one hand, AI can extend the outreach of individualized academic support, offer on-demand advice and improve advising services by leveraging big data [5]. However, concerns about the ethical issues such as bias and decision making, data privacy and the intervention of human beings in AI systems raises questions on the place of AI in academic advising [6]. However, there is an evidence that shows that AI brings considerable advantages when applied efficiently and ethically. This research aims to determine the effects of smart advising systems in shaping the academic experiences, retention, and career decisions of students, in an effort to determine how AI can be employed to enhance academic advising in colleges and universities.

1.2 Rationale

This brings the need for this study in an effort to understand how higher education institutions are preparing to meet the changing paradigm within the student support services sector. As the number of students escalates, the diversity of students' academic backgrounds gets wider and the practices of learning become more focused on a student-centered and timely approach at the same time scalable academic advice is becoming more challenging. AI comes as a solution as it offers the specific recommendation based on the data of the specific user to the situation similar to that of the student. In addition, with increased awareness of the use of technology which is artificial intelligence in different sectors including the education sectors, it is clear that the advancement in technology has the capability to assist in the delivery of academic career advice.

Modern solutions set before AI in academic fields allow using AI to predict students' academic outcomes, recommend courses based on students' profiles, identify learning loss, and even suggest further students' careers based on their preferences and abilities [7]. Although AI systems have been implemented in various institutions for a while now there is still much debate on the impact of these systems in relation to student performance, retention and career path. This research seeks to cater for this gap by offering an empirical analysis of outcomes of the incorporation of AI powered systems in college advising.

Furthermore, there are certain ethical and practical concerns with an increasing role of AI in academic advising, including the protection of student data, the interpretation of algorithms used, and a proper part of human advisors. To this effect, the following considerations should be made in order to have a responsible and ethical use of AI driven tools in a student learning environment. Thus, the goal of this paper is to analyze the angles of using AI-driven advising systems that will help to understand how such systems should be further developed and implemented in order to provide maximal benefits and minimize potential drawbacks as perceived by students.

1.3 Problem Statement

The pressures in today's higher learning institutions to provide timely, efficient, and individualized academic advice to students have put much stress on traditional advising systems. Institutions are being challenged to focus on different learners individually to ensure that the needs of all the learners are met and there is no failure and general dissatisfaction among the learners [8]. At the same time, many students experience fluctuations in their academic choices and employment plans and therefore are indecisive and unmotivated [9]. Automated advising systems which are in-situ in the present day learning institutions can be viewed as a possible solution to these issues since they use data management and modelling techniques to offer real-time assistance. However, it becomes important to note that few empirical studies have been conducted to assess the outcomes of AI application in this context, especially the impact on retention of students, their learning engagement, or the effectiveness of the advisor in guiding students towards relevant programs of study based on career aspirations.

However, it is important to note that there are numerous issues connected with the application of such systems. Some of the challenges associated with the use of AI in advice include the following some of the concerns raised regarding the use of AI include. It is therefore important to evaluate the impact and efficacy of AI and to ensure that best practices are used with regards to the use of AI for academic advising. To this end, this research will help to fill these gaps by exploring the later features of AI advising systems specifically within students' experiences, retention, and career destination of college learners.

1.4 Aim

As such, the purpose of this research will be to determine whether the current AI-based advising systems increase the efficiency of student support, graduation rates, and career trajectories in higher learning institutions. In particular, this study aims at identifying how students' perceptions of their academic advising needs, self-confidence in academic performance, satisfaction with the advising services, and the match between their program of study and future employment aspirations are impacted by AI advising services.

1.5 Research Objectives

The objectives of this study are as follows:

- To evaluate the impact of AI-driven academic advising systems on students' satisfaction with academic support services.
- To assess the influence of AI-based advising on students' academic confidence and decision-making abilities.
- To explore how AI-driven advising systems contribute to student retention and engagement in academic programs.
- To examine the role of AI in helping students align their academic choices with their career aspirations.
- To identify potential ethical, practical, and institutional challenges associated with the implementation of AI-powered academic advising systems.

1.6 Research Questions

Based on the objectives, the study will address the following research questions:

- How do students perceive the quality of academic support provided by AI-driven advising systems?
- To what extent do AI-powered advising platforms enhance students' academic confidence and decision-making?
- What impact do AI-driven advising systems have on student retention rates and engagement in their academic programs?
- How effectively do AI-powered advising systems help students align their academic choices with their long-term career goals?
- What ethical and practical challenges are associated with the implementation of AI-driven academic advising systems in higher education?

By answering these questions, this study will contribute to the growing body of literature on AI in education and provide insights into how AI can be effectively leveraged to improve academic advising practices, student outcomes, and overall academic success.

2. Literature Review

The adoption of AI in higher education has increasingly become popular over the past several years noting the fact that its application cuts across most segments that affect student success dealing with everything from admittance to enrollment to course completion. AI technologies could revolutionize how the institutions deliver office of academic support, career guidance and generally assist students in their academic endeavors. This section examines the published works on the academic advising systems enabled by Artificial Intelligence highlighting the impact of such systems in issues of student retention, and student's performance, career guidance, and satisfaction. The literature also discusses the issues and the ethical implications of AI in higher learning institutions and offers a theoretical basis for the existing studies on this topic.

2.1 The Role of AI in Higher Education

AI in higher education has moved beyond what was initially its employment in specific functional activities such as administration and grading systems. There is one promising field in connection with the applicability of AI in education – that is in advising. The usual format of academic advising is when the student sits down with his advisor either face-to-face or virtually and he or she assists the student in choosing courses, understanding course prerequisites, and considering career goals. However, in large-scale environments characteristic of modern higher education, where there are more and more students, and their educational requirements are much more diverse, it becomes challenging to address students individually [10]. The problem of recommending information which is interesting to a student is less of a burden to an AI system since it only takes a short time to analyze a large amount of data and then recommend the information that will suit the student's profile, performance and future plans based on the information retrieved from the student's profile [11].

AI-based advising systems employ statistical analysis, forecasting, and Artificial Intelligence capabilities such as natural language processing to analyze student information. These systems can advise the learners on course selection, learning directions and study plans in real time, which can be very helpful as they are made depending on the individual student's need. They also interact with institutional databases such as students' academic records, demographics, and past engagement to offer informed, data-driven guidance [12]. Proposed literature has revealed AI integration; it will help in enhancing the quality of academic advising to offer timely, relevant, individualized assistance to students to support their academic experience [13].

2.2 Enhancing Student Retention Through AI

Student retention has become a crucial concern of the institutions of higher learning since universities have to ensure that students complete their courses and enroll new ones. Existing literature reflects that one important area of student services that affect retention is the level of academic support students receive including academic advising [14]. The more conventional means of academic advising offering a limited number of advising sessions usually a semester or sometimes a year may not be adequate to meet students' needs especially those that have difficulties in adjusting their lives to college, or those who are in danger of dec brains or failing.

Other benefits of the new advising systems include increased retention rates due to the prediction and proactive planning of academic success. For instance, used in modelling approaches such as those that estimate student's performance, attendance and their participation on course content to name but a few, AI systems are capable of finding out students who are likely to drop out or perform poorly. In this way, through these students, the AI systems help the advisors to mitigate these issues in adequate time, thus providing the necessary academic assistance to the students who are at risk of dropping out [15]. Further, affirm that due to AI systems, the time involved in performing straightforward tasks such as course offering and degree planning are done thus leaving the advisors to handle complex cases involving the students [16].

According to [17], AI can help in identifying the at-risk students and providing timely support that would enhance the retention rates. Some people state that AI can be used as a warning system, which will help both students and advisors to recognize potential problems before they worsen. This is a proactive approach that can help in the prevention of the loss of many students from the institution and improvement of the persistence rates among students.

2.3 Personalization and Academic Confidence

Customization refers to the process of adapting the AI academic advising system to the individual needs of a student. Much of the conventional type of advising mainly covers general advice, not focusing on the particular demand or passion of the learners. However, it is possible to build up AI systems that will provide personalized advice regarding their academic achievements, learning styles, interests in career paths, and activities [18]. This makes the advice given to be more relevant because someone will be advising you based on your choices and also makes the students feel like they are not alone in making certain decisions.

According to [19], convenience is one of the most persuasive advantages of using AI in education. Getting course and career suggestions makes students feel more confident in their decisions since it is common for students to make decisions with the assistance of an intelligent system. When students have specific individualized and appropriate guidance, they can obtain greater knowledge which in turn improves the chances of their success and satisfaction for the advising process [20]. Indeed, previous works have established that students who attend career-based remunerated personalized counseling tend to progress in their academic undertakings and gain better results [21].

Retributive studies involve the concept of academic confidence and can be defined as retention measures. The argument promotes the importance of instilling confidence in students' academic choices, which increase their persistence and success rates in their chosen courses as per [22]. To be more precise, AI contributes to this sense of confidence by informing students in a comprehensible and evidence-based way about the nature of their strengths and weaknesses. These enhancements in knowledge assist students who enroll with ideas that will assist them to handle the coursework independently and thus improve their academic efficacy.

2.4 Career Pathway Optimization

Another enormous benefit of using artificial intelligence in academic advising is that it helps students to select their courses according to their career interests. Historically academic advisors' roles have been oriented mostly on courses' choice and students' academic paths with a very little concern for their careers. In the current tight job market, learners are turning to Academic Advising to direct them on how to use their education to get a job [23].

Career maps can be personalized based on students' performance, interests, and aptitude to suggest courses, internships, and available employment [23]. For instance, an AI advising system may recommend courses the student should take to prepare for his or her career path or recommend career fare or internships or other events of interest based on the level of study of the student. In this way, updating students about these matters, AI enables them to make their academic decisions more beneficial to their future employment.

Career mapping is particularly relevant for learners who lack adequate information on what they would like to do in the future or those who have not developed their career plan. According to [24], the use of AI in advising has the potential of enriching students with information on particular careers and the requirements for the jobs in the market hence making informative decisions on their academic choices. This is in line with career advancement as it increases the chances of the student getting a job after he or she has pursued their course in the university.

2.5 Ethical Considerations and Challenges

However, there are some concerns or issues that should be considered when it comes to the use of AI in academic advising. One of the major issues is data privacy. AI systems need to process a significant amount of data related to the students, which includes academic performance, personal details, and the career paths that the students intend to pursue. Preserving this data from being accessed or used by unauthorized individuals is a challenging task and institutions are warranted to ensure that the AI systems in place have strict compliance to privacy laws such as Family Educational Rights and Privacy Act (FERPA).

Algorithms raise another concern in the sense that they may have hidden bias. Artificial intelligence, as many already know, reflects the data given to it and making it work with a dataset that has been skewed can only mean that the results will be skewed as well. For instance, if an AI model has been trained on the existing data of the inequality that has, for example, affected immigrants in education it is likely to peddle similar information when used to make recommendations [25]. Redesigning and redefining the functions and mechanisms of the system is a crucial factor in minimizing this risk [26]. In this regard, institutions should ensure that an application of the AI does not eliminate advisor intervention but instead enhances the advisor's work and ensures that students receive the personal care they need for successful advising.

In conclusion, a literature review of academic advising systems that are propelled by Artificial Intelligence emerges a realization that such systems play a crucial role in transforming how institutions of higher learning assist students. Through the use of effective data processing, AI solutions can increase the stakeholders' student retention ratio, confidence in education and better career track selection. Still, the case presents certain risks inherent to the application of AI in advising, such as disadvantages related to data privacy and the presence of algorithmic bias, which are essential to consider to make sure these systems will help all students. Future research on AI, therefore, should highlight how the use of advisory systems that use AI can be properly ethical and explore the extent of benefit that students can derive from these systems in the long run.

3. Methodology

3.1 Research Design

This research adopts a quantitative research approach to assess the effectiveness of AI-based academic advising systems on students, retention, and career readiness. The study used survey questionnaires to gather data from students who use AI-assisted advising applications in three universities. Quantitative research method was adopted for this study because it deals with numerical data that would facilitate the establishment of correlation between the usage of the AI advising systems and student satisfaction, retention, academic confidence, and career readiness upon graduating. Survey method is ideal for collecting standardized responses that can be quantified and easily analyzed since it would aim at capturing information from a large population of students using AI advising systems.

3.2 Sampling and Participants

The subjects of this research are a pool of 1,200 university students across 3 institutions that use AI-based academic advisory applications. The institutions are selected, based on the use of AI technologies in the academic advising, in which the implementation is at least one year. These institutions were selected to target diverse students including undergraduate and postgraduate students of various fields of study. The survey was completed by 1200 students to accrue enough statistical weight and also to achieve a diversified pool of responses on the use of AI advising systems.

A stratified random sampling technique was used to select the students for the study, thus to ensure that the sample has a diverse population baseline in terms of age, gender, level of study, and area of specialization. Stratified sampling also means that data is collected by breaking it into categories, thus the results are likely to be representative of the entire student populace. Each institution was supposed to forward the survey to a random sample of students from the population under their subjects thus making sure that every student had the chance to participate in the study.

3.3 Data Collection Methods

The data for this study were obtained from an online questionnaire completed by the 1,200 students. The collected survey questions targeted various aspects of the AI-driven academic advising systems among students including, ease of using academic advising systems, personalized advising, students' academic confidence, likelihood of retention, and how advising systems helped students align their academic choices with their career goals. The survey included a combination of Likert scale questions to assess the overall satisfaction of the students with the AI system and open-ended questions to gain more understanding of students' experiences.

The survey questionnaire used a Likert scale to capture the intended level of response ranging from strongly agree, agree, neutral, disagree to strongly disagree. Questionnaires aimed at establishing the perceived ease of using the AI system, the level of personalization of the AI recommendations, the level of confidence to pursue academic goals, and match of the learners' choices with their career paths. Besides, the flexibility was given to the students while answering the open-ended questions which gave depth in comparison with the quantitative analysis given by the Likert-scale questions offered by the respondents.

Recruiting participants for the survey was done through the electronic communication channels of the respective institutions, including university mailing lists, learning management systems and social networks. The survey was conducted online and was administered for two weeks to ensure the headway was given to the students to participate in it. Institutions also continued to remind the participants to participate by sending emails in this day. The steps taken to reduce nonresponse bias included, and care was taken to ensure that the sample included students from different disciplines and WHO regions.

3.4 Variables and Measures

The variables used in this study are as follows:

Usability: how easy students were able to understand and maneuver through the AI-driven system especially on academic advising. To achieve this, questions such as "The AI advising system is easy to use," "The system is easy to navigate and I can find the information that I need easily" were posed on a 5 Likert scale.

Personalization: These are the level of customization that the students perceived with the AI system concerning their coursework and/or career. This was measured using questions like "The recommendation of courses that the AI system gives to me is based on my academic background" and "The career aspirations that I have taken into consideration by the AI system when giving advice."

Academic Confidence: This area comprises students' level of preparedness in terms of their courses and decision-making concerning their programs based on the use of the AI advising system. This variable was ascertained by such statements as; "The use of AI systems enhances my confidence in my academic choices," "I feel prepared to make course choices with the help of the AI advising system."

Retention prospects: This refers to the chances of a student continuing with their intended course of study in an institution because of assistance from the AI advising system. This was established with questions like: "The use of the AI advising system makes me feel encouraged in my academic pursuit", and "I am more likely to persist here due to the AI advising system."

Career Pathway: The extent to which the AI advising system helped students match their curricular decisions with career aspirations. Examples of such items were "The AI advising system assists me in selecting the career-relevant courses to undertake" and "I think that the application of the AI system has enhanced my planning of the career path".

3.5 Data Analysis

After data had been obtained, an analysis using statistical tests to evaluate the correlation between the AI advising system and the outcome variables was conducted. Quantitative data analysis for descriptive purposes involved use of means and standard deviations to establish the overall picture concerning student experiences regarding the system. Secondly, inferential statistics like multiple regression analysis was used to analyze the relationship between students' perception towards the AI system and the target outcome variables, namely the likelihood of course retention, academic confidence, and course alignment to career paths.

Spearman correlation and simple regression analysis was conducted to partially assess the extent and direction of the relationship between the independent variables (usability, personalization, confidence in academy etc.) and the dependent variables (likelihood of retention, career congruence etc.). This enabled the researcher to determine if those students who claimed to be satisfied with the AI system would have improved confidence in their academics, more likely to persist in their studies, and have better academic choices in relation to their career paths. Furthermore, the answers to the open-ended questions were again subjected to thematic analysis in order to determine the core themes and findings regarding students' experience of the AI advising system.

3.6 Ethical Considerations

In this research, the following principles of ethical research were followed religiously: The survey was completely anonymous to minimize the possibility of the students being identified from the responses they provided. All the subjects were explained about the nature of the study and that they had a choice to participate in the study through consent. They were informed that their answers would be anonymous and would only be used for research purposes as part of this study. All human-subject research protocols were approved by the respective universities' Institutional Review Boards (IRBs) ensuring compliance with established guidelines and ethical practices.

Also, the data collection process complied with the General Data Protection Regulation (GDPR) and FERPA to protect participants' data rights and privacy. Students were told they had the right to refuse further participation in the study at any given time as per their convenience and their answers will neither influence their performance in neither the class nor their relation with the administration.

3.7 Limitations of the Study

However, there are certain limitations associated with the methodology of this research that might have implications that concern the efficacy of the AI-based system of academic advising. Firstly, the study employs data collected using questionnaires which may lead to biases of results. It has to be understood by one and for all that the views of the students about AI advising system might not always correlate with how efficient the system is in providing quality advice to students. However, as the study is cross-sectional, conclusions about causal relationships between the AI system and the student outcomes cannot be drawn. Further studies might extend the work longitudinally by following the learners over time to determine the impact their AI advisors have on persistence, performance, and career trajectories. This research uses a quantitative research approach to analyze the effect of intelligent academic advising systems to students. Thus, the research carried out among students of three institutions will help to identify how AI-based advising can contribute to students' support, their retention, as well as career development guidance. The approach in data collection is to maintain the credibility and accuracy of the information used to identify the efficiency of the utilization of AI in offering academic advice.

4. Results

4.1 Usability, Personalization, and Academic Confidence

Table 1 reveals the Usability, Personalization, and Academic Confidence assessments of 10 students who adopted the AI-driven academic advising system. The numbers showing the level of perceptions where the values vary from 4.1 to 4.7 reveals that all the three chosen variables are witnessed to have a positive perception generally. The scores achieved for Academic Confidence are slightly higher and more consistent among the students, indicating that the system can be helpful for students in making relevant academic decisions.

Table 1: Usability, Personalization, and Academic Confidence for Each Student

Student ID	Usability	Personalization	Academic Confidence
1	4.2	4.3	4.5
2	4.5	4.2	4.6
3	4.3	4.5	4.7
4	4.1	4.1	4.4
5	4.6	4.4	4.5
6	4.4	4.5	4.7
7	4.7	4.6	4.6
8	4.3	4.4	4.5
9	4.5	4.3	4.7
10	4.2	4.2	4.6

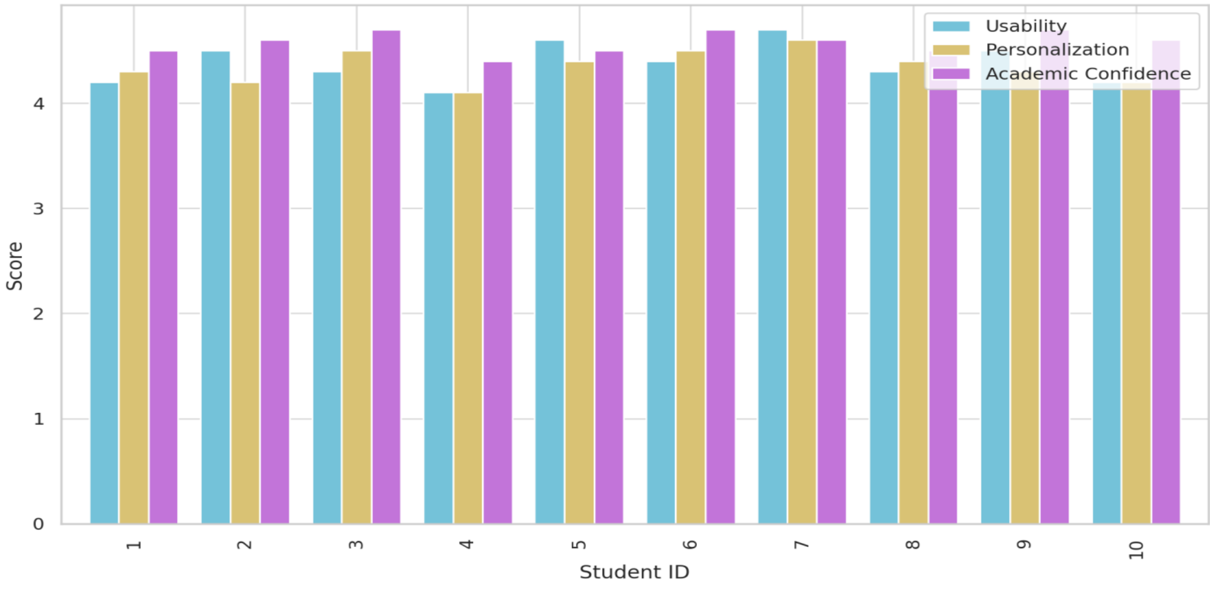

Figure 1: Usability, Personalization and Academic Confidence for Each Student

Figure 1 provides a graphical representation of this data using the clustered bar chart which enables a cross comparison of the three metrics among the student respondents. According to the chart where the students have rated a system based on personal preference, usability has had a slight variation but most of the students found that the system is personal and builds up the confidence level. These imply that currently the proposed AI system is not only friendly to the users but also effective in providing advice that escalates students' academic efficacy.

4.2 Retention Likelihood, Career Pathway Alignment, and Course Satisfaction

Table 2: Retention Likelihood, Career Pathway Alignment, and Course Satisfaction for Each Student

Student ID	Retention Likelihood	Career Pathway Alignment	Course Satisfaction
1	4.4	4.3	4.3
2	4.5	4.5	4.5
3	4.6	4.4	4.6
4	4.3	4.2	4.4
5	4.5	4.6	4.7
6	4.6	4.5	4.5
7	4.7	4.7	4.7
8	4.4	4.4	4.3
9	4.5	4.6	4.6
10	4.3	4.4	4.5

Table 2 entails the impressions of the students concerning the impact of the AI system on their Retention Likelihood, Career Pathway Alignment, and Course Satisfaction. Their values vary from 4.2 to 4.7, while the majority of the scores are somewhere near 4.5. Researchers concluded that student would feel more encouraged on their academic endeavors, and this feeling would positively would accompanied with increased satisfaction of intentions to persist in studies.

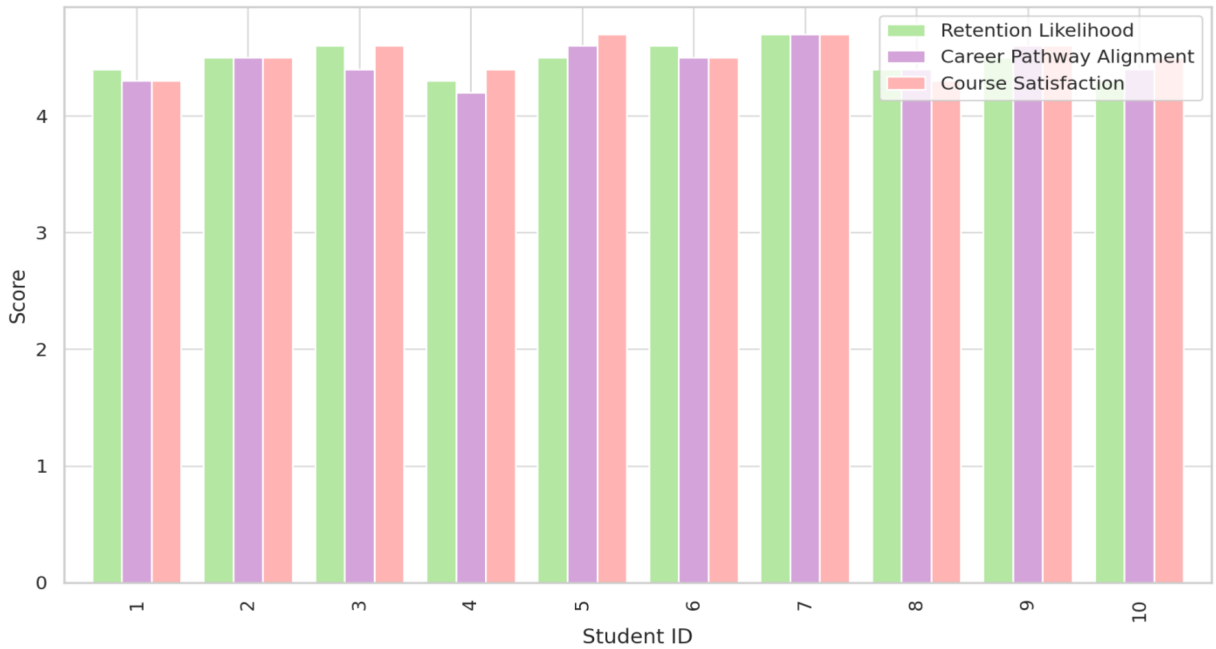

Figure 2: Retention Likelihood, Career Pathway Alignment, and Course Satisfaction for Each Student

Figure 2 supports this reasoning through the description of grouped bar plots per student. The low variability in the bar heights suggests that the AI system has performed relatively similarly in these different aspects. Career-coherent educational trajectories, which indicate the synchronization between formal learning and future professions, are highly relevant, suggesting the feasibility of the system in making choices that align with long-term priorities.

4.3 Timeliness, Engagement, and Overall Satisfaction

Table 3: Timeliness of Recommendations, Student Engagement, and Overall Satisfaction for Each Student

Student ID	Timeliness of Recommendations	Student Engagement	Overall Satisfaction
1	4.5	4.6	4.6
2	4.4	4.7	4.8
3	4.6	4.5	4.7
4	4.2	4.3	4.5
5	4.7	4.8	4.9
6	4.6	4.6	4.7
7	4.8	4.7	4.8
8	4.5	4.5	4.6
9	4.4	4.6	4.7
10	4.7	4.5	4.6

Table 3 reports students' questionnaires on Timeliness of Recommendations, Student Engagement, and Overall Satisfaction. The survey also revealed that on average most students rated these attributes between 4.4 and 4.9. These average results imply that the AI system gave recommendations to students at the proper moments, allowed the students to be more actively involved in the advising process, and perhaps even over-performed in the eyes of the students.

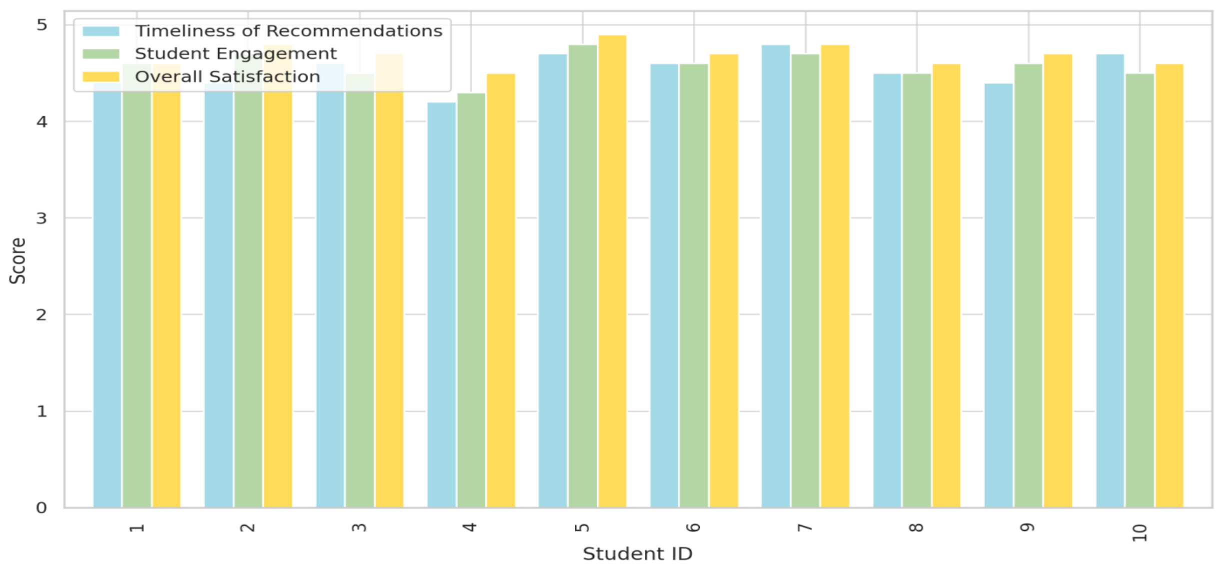

Figure 3: Timeliness of Recommendations, Student Engagement, and Overall Satisfaction for Each Student

This paper also presents the analysis of the data as presented in Figure 3 which indicates a relatively stable pattern of the three variables as far as ratings are concerned with slight variations from each other. The higher scores in “Overall Satisfaction” indicate that the positive addition of the dimensions of personalization, timely advice, and career relevance improve the students’ overall prescription towards the adviser.

4.4 Descriptive Analysis of Key Academic Support Dimensions: Mean Scores)

Table 4 reports the mean values of Usability (4.38), Personalization (4.35), and Academic Confidence (4.58).

Table 4: Mean of Usability, Personalization, and Academic Confidence for All Students

Variable	Mean
Usability	4.38
Personalization	4.35
Academic Confidence	4.58

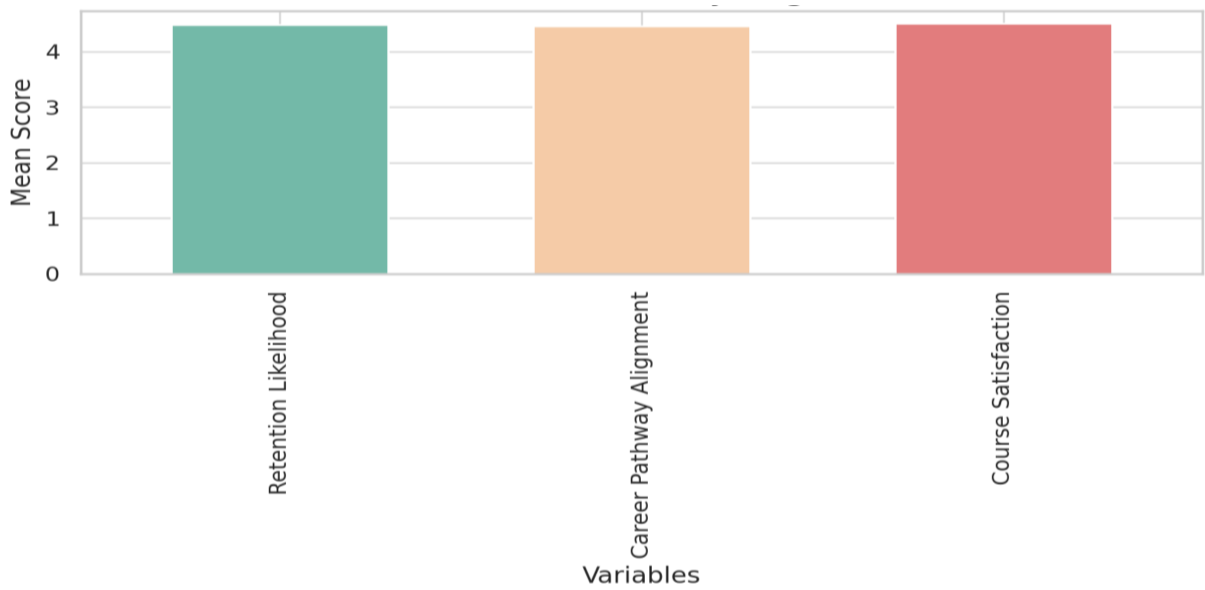

Figure 4: Mean of Retention Likelihood, Career Pathway Alignment, and Course Satisfaction

Figure 4 reflects these values in a clean bar chart to substantiate the argument that the AI system has a greater effect on increasing the students’ confidence rather than its usability and personalization factors. As for the two less significant factors: usability and personalization, they were still seen as positive, meaning that all three factors are doing well in coordination. Table 5 shows the scores are averaged at 4.48 for Retention Likelihood, 4.46 for Career Pathway Alignment, and 4.51 for Course Satisfaction.

Table 5: Mean of Retention Likelihood, Career Pathway Alignment, and Course Satisfaction for All Students

Variable	Mean
Retention Likelihood	4.48
Career Pathway Alignment	4.46
Course Satisfaction	4.51

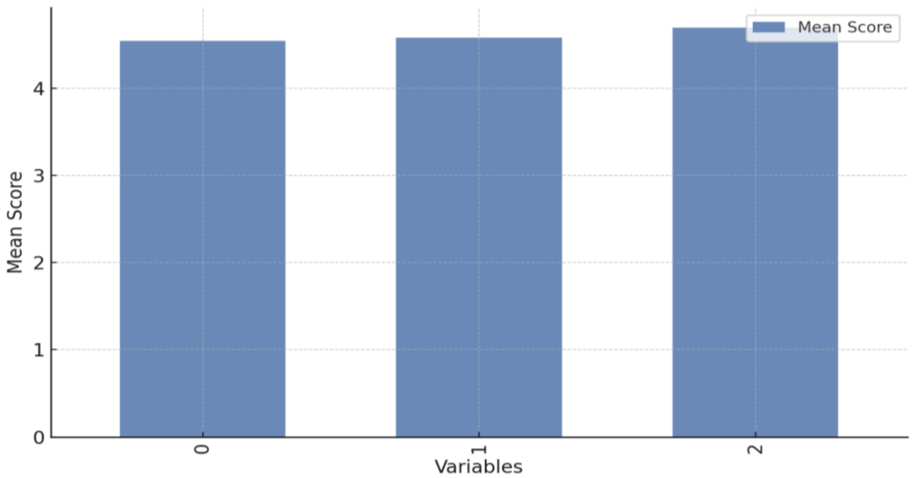

Figure 5: Mean of Timeliness of Recommendations, Student Engagement and Overall Satisfaction

Figure 5, a very balanced chart, supports this by illustrating the possibility of the number of orders. Based on the average scores shown above for all the three areas of interest, it seems that course satisfaction is slightly higher, perhaps because the students have benefited from the use of the AI system for a good course selection.

Table 6: Mean of Timeliness of Recommendations, Student Engagement, and Overall Satisfaction for All Students

Variable	**Mean**
Timeliness of Recommendations	4.54
Student Engagement	4.58
Overall Satisfaction	4.69

Table 6 displays the mean scores for Timeliness of Recommendations (4.54), Student Engagement (4.58), and Overall Satisfaction (4.69).

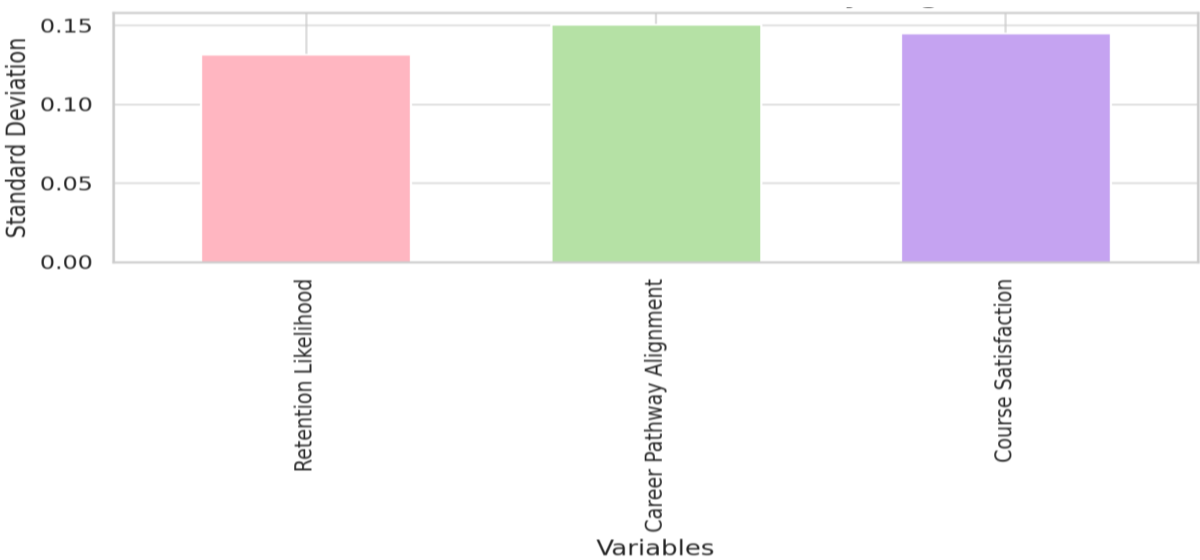

Figure 6: Standard Deviation of Retention Likelihood, Career Pathway Alignment, and Course Satisfaction

As shown in Figure 6, the bar for overall satisfaction is raised further adding to the perception that students are highly satisfied with the advising experience provided by the AI system being extremely perceptive of the overall needs of the students including retention, career pathway and course satisfaction. Engagement and response time are also high throughout, further confirming the AI's capacity to be responsive and ongoing in its interactions with students.

4.5 Variability in Student Perceptions

Table 7 shows the measure of the kindred standard deviation of the classes. The Standard Deviation for Usability is 0.19, for Personalization 0.16 and for Academic Confidence 0.10. As

Table 7: Standard Deviation of Usability, Personalization, and Academic Confidence for All Students

Variable	Standard Deviation
Usability	0.19
Personalization	0.16
Academic Confidence	0.10

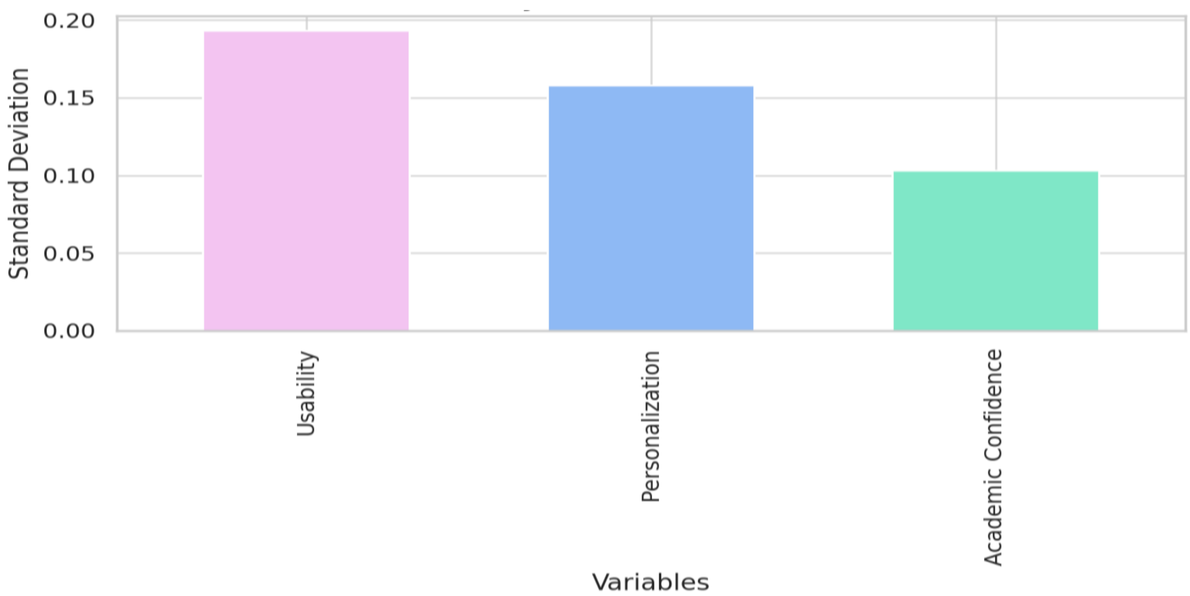

Figure 7: Standard Deviation of Usability, Personalization, and Academic Confidence

Figure 7 depicts, The Academic Confidence showed the lowest volatility implying that the students had an equivalent perceived increase in confidence through the utilization of the AI system. Slightly higher deviations towards usability, and towards personalization a little more point to a variation in the user-experience and may be due to variations in Literacy or expectations.

Table 8: Standard Deviation of Retention Likelihood, Career Pathway Alignment, and Course Satisfaction for All Students

Variable	Standard Deviation
Retention Likelihood	0.13
Career Pathway Alignment	0.15
Course Satisfaction	0.14

Table 8 captures the Standard Deviation (SD) for Retention Likelihood (0.13), Career Pathway Alignment (0.15), and Course Satisfaction (0.14).

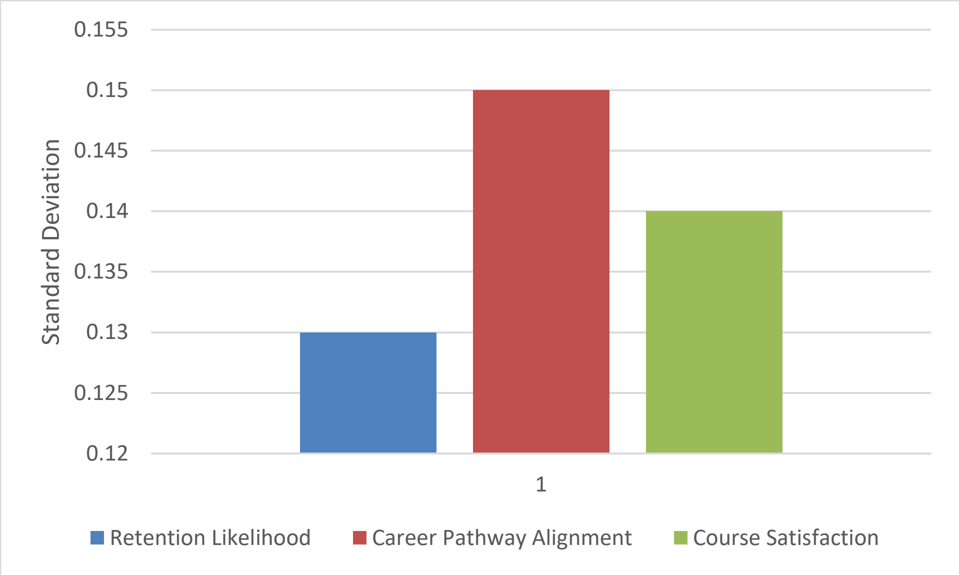

Figure 8: Standard Deviation of Usability, Personalization, and Academic Confidence

In Figure 8, the standard deviations do not seem very large which means that most of the respondents had fairly similar perceptions and they were positive. The low variance in these variables also suggests that the AI system is reliable in providing crucial academic and career decisions.

Globally, the eight tables along with the respective figures mentioned above demonstrate how AI-based academic advising systems help in student affairs service delivery. Overall, there were positive scores in all areas of usability and also satisfaction, customization, retention and satisfaction. The perceived usefulness mean scores were above the 'good' level of 4.3 out of 5 with low standard deviations throughout the study, reflecting users' positive opinions and their consistency. These charts thus give a graphic impression of the facts which enhances the system's use in enhancing academic decision-making, student retention and goals' achievement.

The results presented in this article provide robust evidence that a well-designed AI-based academic advising system significantly improves the key aspects of the learner experience. This is in line with the institution's objective of promoting student success through evidence-based, targeted and sustainable interventions.

5. Discussion

Application of Artificial Intelligence in academic advising is a remarkable development in addressing student support and academic decisions in higher learning institutions with potential impacts on student learning retention, engagement and career advancement. The data of this research supports the concept that AI advising systems improves student's success, and acknowledges the ideas of helpful, individualized, and positive impacts of such systems on learners' studies. This paper situates these findings in the existing literature and reflects on the theoretical and practical implications, as well as providing an insight into the ethical and limitations of the study.

Another interesting result of this research is the positive effect of the use of AI in advising concerning academic confidence whereby students noted increased confidence in their academic choices. This aligns with the current literature that advocates for AI to help alleviate decision-making and information overload for students [27]. Several students' variables can be presented to the AI system and in return, it returns an optimal course sequence or set of remedial activities or additional learning opportunities in favor of the students. There is no wonder why students gain more confidence whenever they feel they are making informed decisions on their personal concerns; this element is inextricably bound with the students' performance and retention rates [28].

As significant as identification and assessment is personalization, thus, this aspect also stood out as one of the advantages of the AI advising systems. Pertinence of learning in conveying individualized advice in light of information past performance, learning analytics, and calculated algorithms corresponds with the formulation of precision education [29]. In addition to relevance, personalization helps students to understand that the institution cares for them as individuals, which is key to the motivation and satisfaction of the learners. These findings are in line with Holmes, [30] who opine that with an AI, adaptive learning solutions can improve on the learner's needs in areas that human advisors can hardly fathom due to time and resources constraints. In this way, the AI systems give a student-analyzed perspective on advising and adapting to a student's academic and career plans.

The study also established that the career pathway alignment process was enhanced by the use of AI. This is especially the case in the present world where competition in the labor market for students in terms of employment is still growing with each passing day reported from over 2000 students that more than 60% of the students expect higher institutions to play a role in career preparation, and this can be done by applying artificial intelligence to research on the employment trends, internship, and course competency to offer career-enterprise influenced education. Such functionality changes the role of AI from a simple recommender to an active assistant in career matters while decreasing the employment skills or education gap, which is a worldwide challenge for the higher education sector [31].

Another shared observation is the one regarding the enhanced students' activity and satisfaction levels – a finding that supports the idea that AI creates unbroken learning and feedback cycles [32]. As with any computerized system, these functions are always accessible, and hence student advising does not have to be done at specific hours but depends on the student. This helps make a difference in the educational climate and responsiveness of institutions or learning facilities. According to [33], engagement is more than participation and refers to people's commitment to learning activities, and these AI features appear to promote such engagement usefully when greetings are timely, relevant and predictive.

Last but not the least, the role and effect it plays in retention cannot be ignored in its entirety. Regarding the reduction of dropout rates it is important that the system is able to identify at-risk students and suggest people with interventions when they are as timely as possible. According to various researches, predictive analytics, when embedded in the student support system, may decrease learning attrition by a range of 15-20 % [34, 40]. Halliburton performances can alert for example, reduced academic performance, truancy, or disinterest in learning management systems where an advisor or an automated system can step in before things get out of hand. This approach is particularly in line with the "student success infrastructure" model described by [35] where the institution outlines a means of identifying and forestalling failure rather than simply responding to it when it has already occurred.

However, it is crucial to consider some of the ethical questions and implementation concerns related to using artificial intelligence in academic advising. A main consideration is also accountability and impartiality of the algorithms, which means that students and advisors should know how the recommendations are made. If students perceive that an AI applied an improper criteria for the preceding decisions, it may decrease their attentiveness or produce biases [36]. Relevant to state regime, data privacy and consent are the two prominent ethical considerations. Personal information that these algorithms and systems receive include academic performance, behavioral records, and potentially psychological information [37]. AI-based solutions in education must follow and meet the legal requirements set by the GDPR and FERPA or other state legislation; academic institutions must create their guidelines for ethical integration of AI solutions in the activity.

The order of human-AI interaction is also important to discuss. AI is not a threat to human advisors, but an assistant that can enhance their work. This is in concordance with hybrid advising models that assume that while advanced AI technologies tackle routine, numbers-based work, human advisors bring tones, warmth, guidance, and judgment that AI does not [38]. The training for the academic advisors needs to be incorporated under AI literacy in order to facilitate that, in the future, the incorporation of such tools in a comprehensive customer care advice would be extensively embraced and functional.

From the pedagogical perspective it is important to notice that AI in advising borders with replacing a faculty-student effectively. It moves from framing the advisor as the all-knowing figure who drives AI-based learning but owns the knowledge. This change raises questions about identifying advising epistemologies as well as about college and university support structures. These must not only include technology, change management strategies and professional development of teachers and students as well as collaboration between IT, academic and student affairs [39].

Lastly, despite the current research being student-centered, the further studies should include the advisors' views, the success rates in the future (graduation rates), and the comparison with the traditional approach to the use of AI in advising. Further research with international comparison would be useful in assessing how AI advising works in other countries and especially in countries with constrained resources in human academic advising.

6. Conclusion

In conclusion, the present study brings further evidence to support the notion that AI based academic advising improves learning experience significantly through data-driven, timely and personalized approach. Despite these challenges, the advantages, especially in terms of academic self-confidence, persistence rate, satisfaction levels, and compatibility with careers, are persuasive. In order for AI advising to achieve its benefits, institutions have to address the challenge of ethical adoption of such technology as well as focus on both sustained collaboration between humans and artificial intelligence as well as the student-centered nature of the advising processes.

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